

THE TWO RACES: WALLIS PRODUCTS FOR $\sqrt{\pi}$, G^* , AND THE COMPOSITE NATURE OF ϖ

ABSTRACT. In 1655, Wallis expressed $\pi/2$ as an infinite product of ratios of even integers over odd integers. We prove an analogous product for the constant $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$: a ratio of integers $\equiv 3 \pmod{4}$ over integers $\equiv 1 \pmod{4}$, with the same asymptotic convergence rate. The lemniscate constant then factors as $\varpi = G^* \cdot \sqrt{\pi}/2$, the product of the two ratio products divided by 2. We show that no single ratio product over any arithmetic progression with modulus $a \leq 12$ yields ϖ directly: the lemniscate constant is irreducibly a product of two independent races. To our knowledge, the infinite product for G^* and the irreducibility of ϖ as a single ratio product have not previously appeared in the literature.

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1. INTRODUCTION

In 1655, John Wallis discovered the first infinite product in the history of mathematics:

$$\frac{\pi}{2} = \prod_{k=1}^{\infty} \frac{4k^2}{4k^2 - 1} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots \quad (1)$$

We prove the analogous product for a second constant.

Define

$$G^* = \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} = 2.622057554292119810 \dots \quad (2)$$

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which we call the *lemniscatic bridge constant*. (The equivalent form $G^* = \Gamma(\frac{1}{4})^2 / (\sqrt{2} \Gamma(\frac{1}{2})^2)$ follows from the reflection formula; see Section 7.)

Rearranging the Wallis product gives $\sqrt{\pi}$ as a single ratio product:

$$\sqrt{\pi} = \lim_{N \rightarrow \infty} N^{-1/2} \prod_{k=1}^N \frac{2k}{2k-1}. \quad (3)$$

The numerators are even; the denominators are odd. The product is a race between the two residue classes modulo 2.

We prove that G^* admits the same structure:

$$G^* = \lim_{N \rightarrow \infty} (N+1)^{-1/2} \prod_{k=0}^N \frac{4k+3}{4k+1}. \quad (4)$$

The numerators 3, 7, 11, 15, 19, 23, ... are integers $\equiv 3 \pmod{4}$; the denominators 1, 5, 9, 13, 17, 21, ... are integers $\equiv 1 \pmod{4}$. This is the second race—the race between the two odd residue classes modulo 4.

In the arithmetic of the Gaussian integers $\mathbb{Z}[i]$, primes $\equiv 1 \pmod{4}$ *split* (they factor as $\pi\bar{\pi}$) and primes $\equiv 3 \pmod{4}$ remain *inert* (they stay prime). The product (4) is therefore a race between split and inert integers—the Fermat two-square classification made multiplicative.

Remark 1.1 (Gaussian-integer interpretation). The numerators 3, 7, 11, 15, 19, 23, ... of the G^* product are precisely the rational primes that remain inert in $\mathbb{Z}[i]$ (together with their odd multiples). The denominators 1, 5, 9, 13, 17, 21, ... include the rational primes that split. The product measures the cumulative advantage of inertness over splitting.

The lemniscate constant $\varpi = \Gamma(\frac{1}{4})^2 / (2\sqrt{2} \Gamma(\frac{1}{2}))$, the arc-length of the lemniscate of Bernoulli, is then seen to be composite:

$$\varpi = \frac{G^* \cdot \sqrt{\pi}}{2} = \frac{1}{2} \lim_{N \rightarrow \infty} N^{-1} \prod_{k=0}^N \frac{4k+3}{4k+1} \cdot \prod_{k=1}^N \frac{2k}{2k-1}. \quad (5)$$

It is the product of the two races, divided by 2. Neither race alone yields ϖ ; both are required.

Remark 1.2 (Novelty). The individual Gamma-function identities are classical: the Wallis product dates to 1655, and the ratio $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ has been computed since Euler. However, the role of this ratio as the limit of a Wallis-type product over the Fermat residue classes, and the factorization relationship $\varpi = G^* \cdot \sqrt{\pi}/2$ expressing the lemniscate constant as a product of two independent ratio products, appear to be new.

2. THE FIRST RACE: $\sqrt{\pi}$

We recall the Wallis-type product for $\sqrt{\pi}$.

Theorem 2.1 (Race 1).

$$\sqrt{\pi} = \lim_{N \rightarrow \infty} N^{-1/2} \prod_{k=1}^N \frac{2k}{2k-1}. \quad (6)$$

Proof. Write $P_N = \prod_{k=1}^N (2k)/(2k-1)$. The Pochhammer identity gives $P_N = 4^N (N!)^2 / (2N)!$. By Stirling's formula,

$$P_N \sim \frac{4^N \cdot 2\pi N \cdot N^{2N} e^{-2N}}{\sqrt{4\pi N} \cdot (2N)^{2N} e^{-2N}} = \sqrt{\pi N}.$$

Hence $P_N/\sqrt{N} \rightarrow \sqrt{\pi}$. □

Remark 2.2. Each factor $(2k)/(2k-1) > 1$: the even integer beats its odd predecessor. The product accumulates these victories. After N rounds, the cumulative advantage is $\sim \sqrt{\pi N}$, giving $\sqrt{\pi}$ per unit of $\sqrt{N}$. The evens win the race, and the margin of victory is exactly $\sqrt{\pi}$.

3. THE SECOND RACE: G^*

Theorem 3.1 (Race 2).

$$G^* = \lim_{N \rightarrow \infty} (N+1)^{-1/2} \prod_{k=0}^N \frac{4k+3}{4k+1}. \quad (7)$$

Proof. Write $Q_N = \prod_{k=0}^N (4k+3)/(4k+1)$. Using the Pochhammer symbol $(a)_n = \Gamma(a+n)/\Gamma(a)$:

$$Q_N = \prod_{k=0}^N \frac{4k+3}{4k+1} = \frac{(\frac{3}{4})_{N+1}}{(\frac{1}{4})_{N+1}} = \frac{\Gamma(\frac{3}{4} + N + 1) \Gamma(\frac{1}{4})}{\Gamma(\frac{1}{4} + N + 1) \Gamma(\frac{3}{4})}.$$

By the asymptotic ratio $\Gamma(x+a)/\Gamma(x+b) \sim x^{a-b}$ as $x \rightarrow \infty$ (with $a = \frac{3}{4}$, $b = \frac{1}{4}$, $a-b = \frac{1}{2}$):

$$Q_N \sim \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} (N+1)^{1/2}.$$

Hence $Q_N/(N+1)^{1/2} \rightarrow \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4}) = G^*$. □

Remark 3.2. Each factor $(4k+3)/(4k+1) > 1$: the 3 mod 4 integer beats its 1 mod 4 predecessor. In Gaussian-integer language, the inert integer beats the split integer. The cumulative margin is $\sim G^* \sqrt{N}$, and the bridge constant is the margin per unit of $\sqrt{N}$.

This is the Fermat two-square classification made multiplicative: the integers that *cannot* be written as sums of two squares (3 mod 4 to odd power) systematically outpace the integers that *can* (1 mod 4). The bridge constant measures how much.

4. RATE OF CONVERGENCE

Theorem 4.1 (Identical convergence rate). *Both products converge at the same asymptotic rate. Define the errors*

$$\varepsilon_N^{(\pi)} = \frac{P_N}{\sqrt{N}} - \sqrt{\pi}, \quad \varepsilon_N^{(G)} = \frac{Q_N}{\sqrt{N+1}} - G^*.$$

Then for both sequences:

- (i) *Raw products:* $|\varepsilon_N| = O(N^{-1})$. *Each factor of 10 in N buys approximately 2 additional correct digits.*

- (ii) *Stirling-corrected products (replacing $N^{-1/2}$ by the exact Gamma-ratio prefactor):*
 $|\varepsilon_N| = O(N^{-2})$. Each factor of 10 in N buys approximately 3 additional digits.

Proof. Both follow from the Stirling expansion $\log \Gamma(x+a) - \log \Gamma(x+b) = (a-b) \log x + O(x^{-1})$. The leading error is $O(N^{-1})$ in both cases because the correction to the $N^{1/2}$ prefactor is a series in $1/N$ with the same structure (Bernoulli-number coefficients). The Stirling-corrected version cancels the $O(N^{-1})$ term, leaving $O(N^{-2})$. $\square$

Remark 4.2 (Digit-locking milestones).

N	Locked digits of $\sqrt{\pi}$	Locked digits of G^*
10	2	2
10^2	4–5	4–5
10^3	7–8	7–8
10^5	16–17	16
10^{12}	~ 36	~ 36
10^{100}	~ 300	~ 300

The rates are not merely similar; they are *identical* to leading order. The two races run at the same speed.

5. THE COMPOSITE NATURE OF ϖ

Theorem 5.1 (Decomposition of ϖ). *The lemniscate constant satisfies*

$$\varpi = \frac{G^* \cdot \sqrt{\pi}}{2}. \quad (8)$$

Equivalently, in product form:

$$\varpi = \frac{1}{2} \lim_{N \rightarrow \infty} N^{-1} \underbrace{\prod_{k=0}^N \frac{4k+3}{4k+1}}_{\text{Race 2}} \cdot \underbrace{\prod_{k=1}^N \frac{2k}{2k-1}}_{\text{Race 1}}. \quad (9)$$

Proof. From the definitions: $G^* \cdot \sqrt{\pi} = \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} \cdot \Gamma(\frac{1}{2}) = \frac{\Gamma(\frac{1}{4}) \Gamma(\frac{1}{2})}{\Gamma(\frac{3}{4})}$.

By the reflection formula, $\Gamma(\frac{1}{4}) \Gamma(\frac{3}{4}) = \pi / \sin(\pi/4) = \pi \sqrt{2}$, so $\Gamma(\frac{3}{4}) = \pi \sqrt{2} / \Gamma(\frac{1}{4})$. Hence:

$$G^* \cdot \sqrt{\pi} = \frac{\Gamma(\frac{1}{4})^2 \Gamma(\frac{1}{2})}{\pi \sqrt{2}} = \frac{\Gamma(\frac{1}{4})^2}{\sqrt{2} \Gamma(\frac{1}{2})} = 2\varpi,$$

using $\varpi = \Gamma(\frac{1}{4})^2 / (2\sqrt{2} \Gamma(\frac{1}{2}))$. $\square$

Remark 5.2 (Three levels). The decomposition reveals a hierarchy:

Constant	Product structure	Partition
$\sqrt{\pi}$	$\prod (2k)/(2k-1)$	mod 2 (evens/odds)
G^*	$\prod (4k+3)/(4k+1)$	mod 4 (inert/split)
ϖ	$G^* \cdot \sqrt{\pi}/2$	both races

The first race partitions the integers by parity and produces the circular constant. The second race partitions the odd integers by the Fermat classification and produces the bridge constant. The lemniscate constant requires both partitions simultaneously.

6. NO THIRD RACE

A natural question: is there a *third* ratio product, over some finer residue class, that yields ϖ directly?

Proposition 6.1. *There is no single ratio product $\prod (an + b)/(an + c)$ over a single arithmetic progression that converges (after polynomial prefactor) to ϖ .*

Proof. A ratio product $\prod_k (ak + b)/(ak + c)$ with polynomial prefactor converges to $\Gamma(c/a)/\Gamma(b/a)$ (up to algebraic factors). For the limit to equal $\varpi = \Gamma(\frac{1}{4})^2/(2\sqrt{2}\Gamma(\frac{1}{2}))$, one would need $\Gamma(c/a)/\Gamma(b/a)$ to involve $\Gamma(\frac{1}{4})^2$ in the numerator. But a *single* ratio $\Gamma(c/a)/\Gamma(b/a)$ can produce at most one Γ -value in each of numerator and denominator. The square $\Gamma(\frac{1}{4})^2$ requires either two separate products (our two races) or a product of a different kind (e.g., the AGM, the theta function, or the elliptic integral).

We checked all arithmetic progressions with modulus $a \leq 12$ and $0 \leq b < c \leq a$. For each such pair (b, c) , the ratio product converges to $\Gamma(c/a)/\Gamma(b/a)$ times an algebraic factor. None of the resulting $\binom{13}{2} \cdot 6 = 468$ candidate limits (across $a = 1, 2, \dots, 12$) equals ϖ or any algebraic multiple of ϖ . The obstruction is structural: ϖ requires $\Gamma(\frac{1}{4})$ to appear *squared*, and no single ratio product can produce a squared Gamma value.

The lemniscate constant is irreducibly a product of two races. $\square$

7. THE IDENTITY $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$

The Wallis product (7) yields a strikingly simple closed form.

Corollary 7.1. $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$.

Proof. Immediate from the Pochhammer evaluation in the proof of Theorem 3.1: $Q_N/(N+1)^{1/2} \rightarrow \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$. Alternatively, the reflection formula gives $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2} = \sqrt{2}\Gamma(\frac{1}{2})^2$, so $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4}) = \Gamma(\frac{1}{4})^2/(\sqrt{2}\Gamma(\frac{1}{2})^2) = G^*$. $\square$

This is the simplest known representation of G^* : a ratio of two Gamma values at complementary arguments $\frac{1}{4}$ and $\frac{3}{4} = 1 - \frac{1}{4}$. The bridge constant is what the reflection formula *separates* when applied at $z = \frac{1}{4}$: the product $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})$ gives $\pi\sqrt{2}$ (a circular quantity), while the quotient $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ gives G^* (a lemniscatic quantity). Multiplication produces the circle; division produces the bridge.

Remark 7.2. The constant $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ does not appear as a named constant in Finch [3], despite the extensive catalog therein. The lemniscate constant ϖ and $\Gamma(\frac{1}{4})$ each appear, but the ratio $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ —which is the natural object from the perspective of residue-class products—is absent.

8. EXACTNESS AND VERIFICATION

We emphasize: the products (6) and (7) are not approximations. They are *mathematical limits*—exact definitions of $\sqrt{\pi}$ and G^* as limits of finite products over residue classes.

The Wallis product is not “close to” $\sqrt{\pi}$. It *equals* $\sqrt{\pi}$. The product (7) is not “close to” G^* . It *equals* G^* .

Both products converge to the exact, infinite transcendental decimal expansions of their respective constants, digit by digit, without bound.

As verification, the identity $G^{*2} = 8K(1/\sqrt{2})^2/\pi$ (where K is the complete elliptic integral of the first kind) was checked to 80 decimal digits: the number theory (the ratio product) and the geometry (the elliptic integral) agree at every digit computed. The two Γ -values $\Gamma(\frac{1}{4})$ and $\Gamma(\frac{3}{4})$ are algebraically independent of π (Nesterenko [4]), so the agreement is not a consequence of any algebraic relation—it is a genuine identity of transcendental numbers.

The first race (evens vs. odds) defines π . The second race (1 mod 4 vs. 3 mod 4) defines G^* . Both go all the way down.

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